Hui "Amalie" Shi

Oxford, UK | amalie.hui.shi@gmail.com | linkedin.com/in/amalieshi | amalieshi.github.io | ORCID: 0009-0001-2931-1414

PROFESSIONAL SUMMARY

Experienced biomedical and software engineer specialising in healthcare automation and medical imaging verification. With degrees from Johns Hopkins University and Georgia Tech, integrates Python-based frameworks to improve clinical software reliability. Certified PMP, EIT, ISTQB CTAL-TAE, and ASQ CSSYB — advancing innovation in digital healthcare systems across pharmaceutical and medical device industries.

INDUSTRIAL EXPERIENCE

Software Test Engineer | Adaptix Ltd. *Oct 2022 – Present*

- Architected and deployed a suite of 30+ enterprise-grade automation frameworks using Django, PyTest, and PyWinAuto to verify 3D medical imaging UIs and communication portals; modernised verification workflows and significantly increased test coverage for high-stakes, mission-critical healthcare systems.
- Led end-to-end validation and verification of design requirements for veterinary and orthopaedic X-ray systems, ensuring compliance with IEC 62304 and other international regulatory standards.
- Authored and presented detailed technical test reports to cross-functional teams, enabling resolution of 45+ software and hardware defects through analytical troubleshooting and collaborative problem-solving.
- Streamlined testing workflows and quality processes, resulting in improved defect detection efficiency and faster release cycles.

Software Applications Engineer | Informetric Systems Inc. *Feb 2021 – Aug 2022*

- Gathered, analysed, and mapped user and process requirements into configurable automation logic for InfoBatch manufacturing systems, driving digital transformation and process optimisation for pharmaceutical clients.
- Developed, tested, and validated 5 software updates and hotfixes for InfoBatch; managed the software release and product launch of InfoLog, including VM-based test environments, installation packaging (Advanced Installer), and automated test procedures.
- Provided high-level technical support to resolve complex issues — HTML report formatting, delayed data uploads, printer integration — ensuring system uptime and customer satisfaction.
- Collaborated with software developers, QA, and client engineering teams to align requirements and deliver robust, production-ready solutions.

Pharmaceutical Engineer | Aphena Pharma Solutions *Jan 2020 – Feb 2021*

- Authored and executed 30+ validation protocols and reports covering equipment validation (IQ/OQ/PQ) and process validation (PV/PPQ), releasing manufacturing batches and commissioning 8+ automated systems, including blending tanks, labelling machines, water purification units, and serialisation equipment, under FDA cGMP standards.
- Designed and optimised 5+ manufacturing and lab-scale processes using engineering design principles and data-driven analysis, improving process efficiency and product consistency.
- Implemented CAPA and preventive maintenance programmes, driving equipment reliability, reduced downtime, and sustained cost efficiency across production operations.
- Collaborated with R&D, QA, and manufacturing teams to deliver engineering process improvements, ensuring ongoing compliance with FDA regulatory and technical standards.

PROJECTS

DICOM Conformance Verification of the Data Exchange Platform *2024 – Present*

Designed and delivered an end-to-end automated conformance verification pipeline for a healthcare Data Exchange Platform:

- Validated DICOM service compliance across four critical healthcare workflows — Modality Worklist (MWL), Modality Performed Procedure Step (MPPS), Radiation Dose Structured Report (RDSR), and DICOM Storage — using the DICOM Validation Toolkit (DVTK).
- Developed a full-stack web application to orchestrate DVTK validation runs, aggregate conformance results, and surface structured reports, replacing ad hoc manual checks with a repeatable, traceable verification workflow.
- Improved auditability and regulatory traceability by consolidating conformance evidence into a single platform, enabling faster identification of protocol deviations across system configurations.

Workflow-Based Automation of UI Software Testing — *2022-2024*

Engineered an automated testing framework for medical imaging software UI, improving safety verification of a compact, low-dose 3D X-ray imaging system (Adaptix):

- Built a PyWinAuto-based automation framework simulating end-user behaviour at scale, executing image acquisition workflows thousands of times without manual intervention.
- Reduced manual testing effort and human error by integrating continuous verification into the software development lifecycle, enabling early detection of regressions and improving software robustness.

InfoBatch Software Upgrade & Validation — *2021-2022*

Led end-to-end execution and validation of a multi-site InfoBatch software upgrade programme on AWS cloud infrastructure:

- Biogen: Managed a two-site upgrade from v3.0 to v4.0, ensuring continuity of data integrity and system functionality post-migration.
- RE Mason: Executed a two-site upgrade from v2.0 to v4.0, spanning a broader version delta requiring additional regression validation.

SKILLS

Engineering & Development: Python, Django, PyTest, PyWinAuto, DVTK, HTML, CSS, Tailwind, JavaScript, Git, MATLAB, Microsoft SQL Server

Data & Analysis: Statistical Analysis, Data Mining, Data Management, Reporting & Visualisation, R, Database Management

Regulatory & Quality: IEC 62304, FDA cGMP, IQ/OQ/PQ Validation, CAPA, DICOM, Medical Device Software Verification & Validation

Management & Collaboration: Project Management (PMP), Jira, Process Improvement, Product Requirement Management, Cross-functional Team Leadership

EDUCATION

M.S. Biomedical Engineering | Johns Hopkins University, Whiting School of Engineering — *Dec 2019*

Thesis: The Use of 3D-Printed Phantoms for Evaluating CT Image Quality

B.S. Biomedical Engineering (Highest Honors) | Georgia Institute of Technology — *Dec 2016*

Thesis: The Use of Implanted Intramuscular FES System for Ameliorating Foot Drop During Locomotion

PROFESSIONAL CERTIFICATIONS

ISTQB Certified Test Automation Engineer (CTAL-TAE)	BCS / Chartered Institute for IT	May 2025
Project Management Professional (PMP)	Project Management Institute (PMI)	Feb 2025
Cloud Web Development Bootcamp	HyperionDev	Dec 2024
ISTQB Certified Foundation Level 4.0	BCS / Chartered Institute for IT	Nov 2024
Engineer-In-Training (EIT) — Chemical Engineering	Maryland State Board for Professional Engineers	Aug 2021
Certified Six Sigma Yellow Belt (CSSYB)	American Society for Quality (ASQ)	Sep 2020

PUBLICATIONS & CONFERENCES

Peer-Reviewed Journal & Conference Papers

[1] Shi, H., Woods, D., Egan, C., & Wilson, A. (2026, May). Building scalable and automated verification tools for healthcare imaging systems. *Medical Informatics Europe (MIE 2026)*, Genova, Italy. PubMed

[2] Shi, H., Gang, G., Li, J., Liapi, E., Abbey, C., & Stayman, J. W. (2020). Performance assessment of texture reproduction in high-resolution CT. *Proceedings of SPIE Medical Imaging*, Vol. 11316. PMC

[3] Shi, H., Lyle, M., Turtill, C., & Nichols, R. (2017). Positive force feedback may ameliorate muscle weakness. *Society for Neuroscience 47th Annual Meeting (SfN 2017)*, Washington, D.C. Abstract

Invited Talks & Conference Presentations

[4] Shi, H. (2025, March). The ultimate Swiss army knife of UI software testing: Accelerating workflow-based automation [Conference talk]. *Selenium Conference 2025*, Valencia, Spain. Recording

[5] Shi, H. (2024, July). Presenting a workflow of automating UI software testing and reviewing a commercial case study [Conference presentation]. *RSLondon Southeast 2024*, Imperial College London. Abstract

Theses

[6] Shi, H. (2020). *The use of 3D-printed phantoms for evaluating CT image quality* [Master's thesis, Johns Hopkins University]. JScholarship. Repository

[7] Shi, H. (2017). *The use of implanted intramuscular FES system for ameliorating foot drop during locomotion* [Undergraduate thesis, Georgia Institute of Technology]. Georgia Tech Repository. Repository